Heidelberg Hackathon

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1 Introduction

Automatic cell segmentation in microscopic images is critical for quantitative biological analysis, enabling high-throughput studies of cell morphology and dynamics. This project compares advanced U-Net-based convolutional neural networks (CNNs) to optimize segmentation accuracy across three imaging modalities: fluorescence, differential interference contrast (DIC), and phase-contrast microscopy. Key challenges include low contrast in DIC images, halo artifacts in phase-contrast, and variable staining intensity in fluorescence data.

2 Data Augmentation

We applied the following augmentation pipeline during training with an execution probability of 0.1 per transform. Geometric transforms include random 90° rotations and elastic deformations, while photometric adjustments cover contrast and brightness variations. Label transforms mirror geometric changes with nearest-neighbor interpolation for masks.

Figure 1

3 Architectural Variants

3.1 U-Net Baseline

The baseline U-Net [1] uses a contracting path (4 downsampling blocks) and an expansive path with skip connections. Our implementation adds batch normalization and deeper convolutional blocks (2 extra layers per stage) compared to the original architecture [2]. We use zero-padding (SAME padding) in the convolution layers to ensure that the output size of the network is the same as the input size. This eliminates the need for cropping and resizing of the images in the original model.

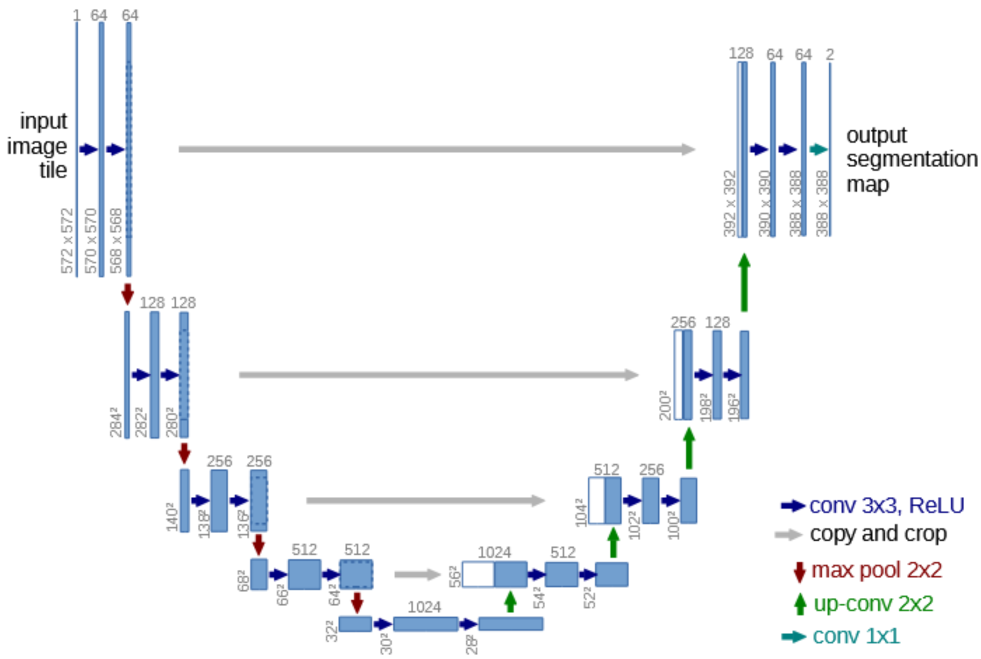


Figure 2: U-Net architecture

3.2 UNet++

UNet++ [3] introduces nested dense skip connections between encoder and decoder sub-networks, enabling multi-scale feature fusion through convolutional layers on skip pathways.

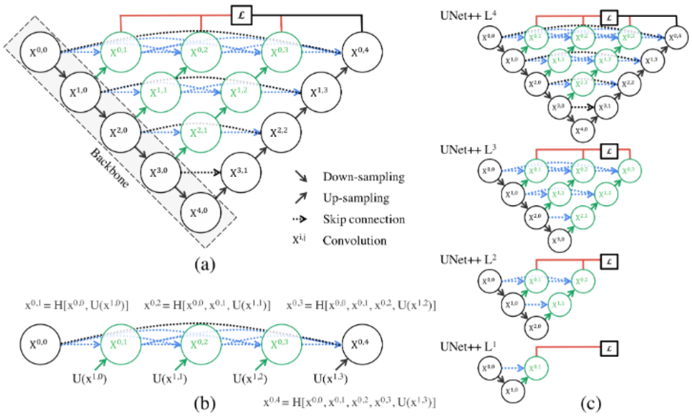


Figure 3: U-Net++ as improvement

3.3 Attention U-Net

Incorporates squeeze-and-excitation attention gates [4] prior to skip concatenation. Gates learn spatial attention maps $\alpha \in [0,1]^{H \times W}$ to emphasize cell boundaries:

$$\alpha_{i,j} = \sigma(\mathbf{W}_a[\mathbf{f}_{enc}; \mathbf{f}_{dec}] + \mathbf{b}_a) \tag{1}$$

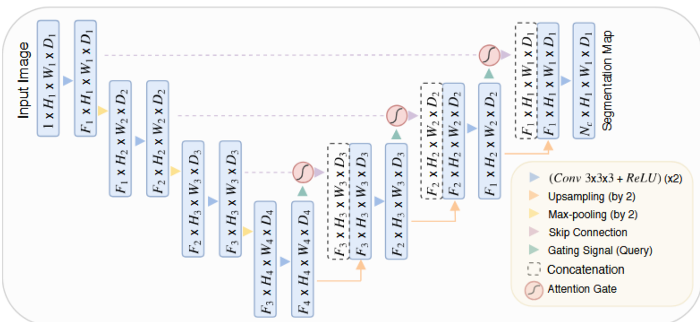


Figure 4: Attention mechanism in U-Net

3.4 RRCNN U-Net

Recurrent Residual CNN (R2U-Net) [6] replaces standard convolutional layers with recurrent blocks (2 GRU steps) and residual connections. Our RRCNN variant adds channel attention [7] to recurrent units.

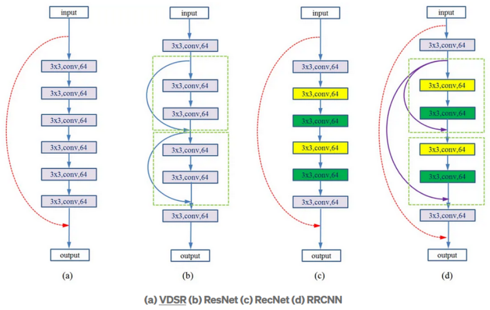


Figure 5: Recursive Residual CNN

3.5 BiFPN U-Net *(In Progress...)*

Integrates Bidirectional Feature Pyramid Network [5] as the decoder. Features from higher levels are fused using learnable weights w_i normalized via fast softmax:

$$w_i = \frac{e^{\gamma \cdot w_i}}{\sum_j e^{\gamma \cdot w_j}}, \quad \gamma \geq 1 \tag{2}$$

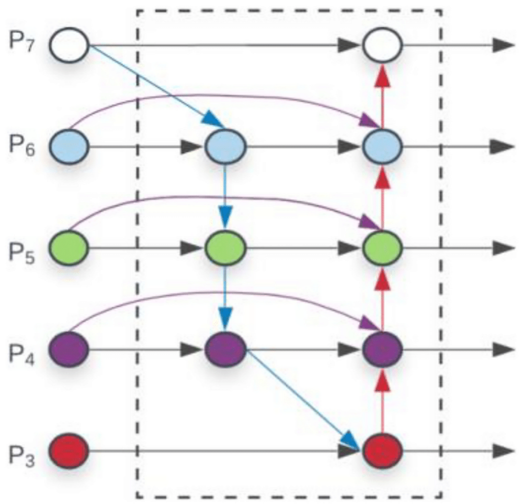


Figure 6: Weighted Bi-directional Feature Pyramid Network

4 Experiments

4.1 Implementation Details

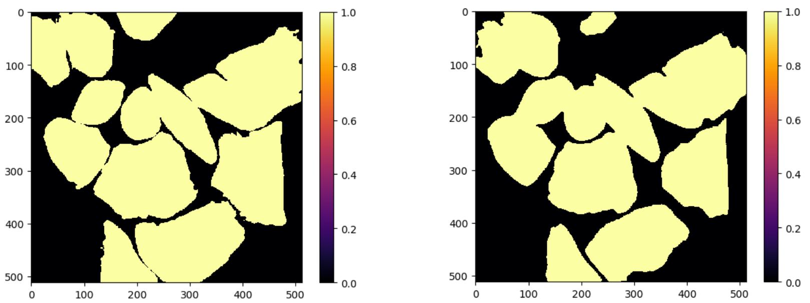
All models trained for 64 epochs using Adam ($\beta_1 = 0.9$, $\beta_2 = 0.999$) with LogCoshDiceLoss loss. Initial LR is for now 1e-3 .

4.2 Results

Table 1 shows segmentation performance (Dice) on the HeLa test set. U-Net++ achieves the best overall Mean Intersection over Union (mIoU). From practice, it appears that the best loss function is LogCoshDiceLoss, but we have not yet tested the weighted average of Dice Loss and Binary Cross Entropy.

Model	DIC	Fluorescence	Phase-contrast	All
U-Net	0.855	?	?	?
UNet++	0.867	?	?	?
Attention U-Net	0.835	?	?	?
RRCNN U-Net	0.810	?	?	?
BiFPN U-Net	?	?	?	?

Table 1: LogCoshDiceLoss across imaging modalities.



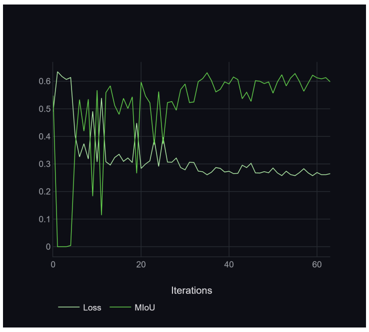
Mask in validation dataset | Example model output

Figure 7: The prediction of the model generally follows the markings, but whatever method is adopted, the biggest problem for the model is the inclusion of the background between the marked objects.

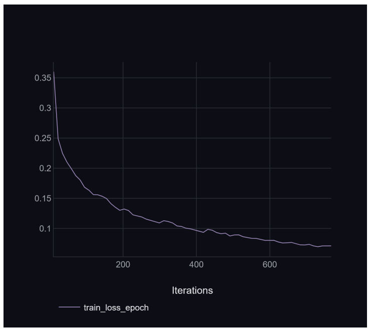
5 Conclusion

Our analysis demonstrates:

- Setting the dropout in the network, even at 0.1, results in significantly less overfitting
- Extended U-Net architectures have the potential for a higher score, but in all likelihood it will not be significantly better.
- We don't understand exactly why, but using google colab, our models perform noticeably worse, the final experiments will be performed on virtual machines that we have recently been given access to.
- Due to the use of silver-truth rather than golden-truth data (the method brings with it additional noise), it also seems important to use different data augmentation, which may ultimately affect the results obtained by the best model.
- The biggest problem of the model is the consideration of the background between cells, if we have enough time we will try to increase the weight that the model places on predicting the boundary pixels. The problem is that the data, due to the acquisition methodologies, already has a dose of noise with it (we don't use golden-truth data, their addition causes the prediction to drop because they are however significantly different from each other).

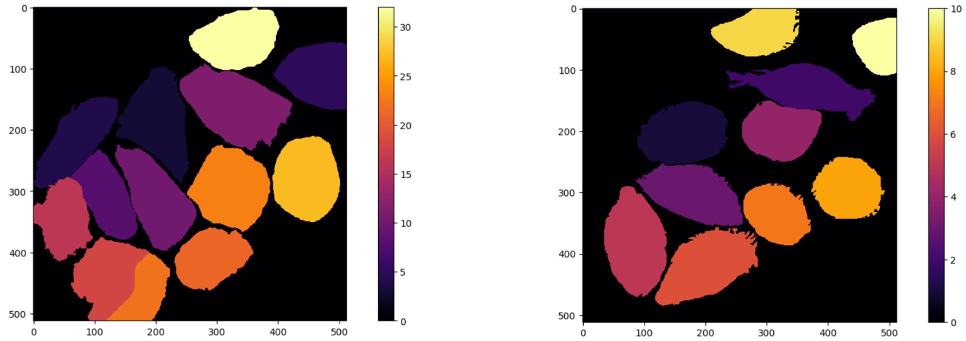


MIoU on Validation dataset



Training loss

Figure 8: Example of model training (Google Colab) using ClearML to effectively compare results.



Silver-Truth   Golden-Truth

Figure 9: Comparison beetwen silver and golden truth for the same image.

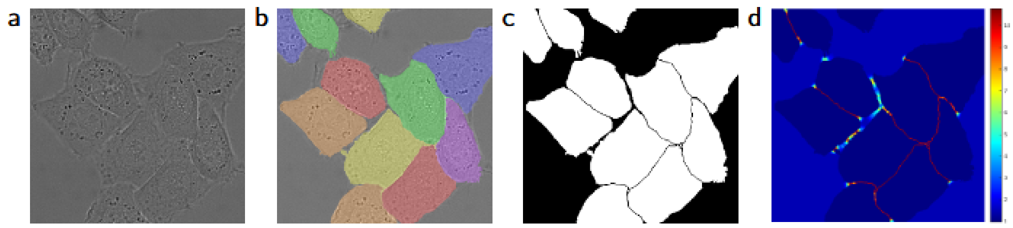


Figure 10: Possible improvement of the model to pay more attention to the pixels that are the boundaries of a given cell.

References

[1] O. Ronneberger, P. Fischer, and T. Brox, "U-Net: Convolutional Networks for Biomedical Image Segmentation," in *MICCAI*, 2015.

[2] T. Falk *et al.*, "U-Net: deep learning for cell counting, detection, and morphometry," *Nature Methods*, 2019.

[3] Z. Zhou *et al.*, "UNet++: A Nested U-Net Architecture for Medical Image Segmentation," *arXiv:1807.10165*, 2018.

[4] O. Oktay *et al.*, "Attention U-Net: Learning Where to Look for the Pancreas," *arXiv:1804.03999*, 2018.

[5] T. Chen *et al.*, "Efficient and Accurate 2D Feature Fusion with BiFPN," *arXiv:2004.00375*, 2020.

[6] M. Z. Alom *et al.*, "Recurrent Residual U-Net for Medical Image Segmentation," *arXiv:1802.06955*, 2018.

[7] G. Wang *et al.*, "RRCNN-Unet: R2U-Net with Recurrent Convolutional Layers," *IEEE TMI*, 2020.